Who Bore Britain's Trade Shocks? A Household Anatomy of the Post-Covid Episodes*

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August 2026 — working draft

Abstract

Britain's post-Covid decade delivered three trade-policy shocks—the TCA border (2021), the US tariffs (2025), the India agreement (2026)—plus the sharpest trade-transmitted terms-of-trade shock since the 1970s (2022) and a near-zero benchmark (CPTPP). This paper studies them under one household lens: each episode a declared first stage from published estimates or government projections, mapped onto households through prices and earnings, compared in ratios, with the policy/market distinction explicit. First-pass results: the negative shocks were regressive (the gross energy shock would have cost the bottom decile 12.39 per cent of spending versus 5.77 at the top, before policy and demand response; TCA food 0.35 versus 0.20 on the window-average path), the projected positive gains are proportional and negligible, and the cushioning was margin-specific: automatic on earnings (36.6–43.9 per cent), discretionary on prices (the Energy Price Guarantee absorbed 46.2 per cent of the gross shock), with indexation a year late—£230 to £393 for a single adult's Universal Credit standard allowance in the crisis year, by convention. Two bounds are carried throughout: decile gradients are real but capture a small share of the variance in who was hurt, and the consumption-basket channel priced here is smaller than the net-nominal-position and labour-income channels it omits. A PolicyEngine microsimulation stage, specified against verified model capabilities, addresses the first and completes the study.

Keywords: trade shocks, tariffs, Brexit, terms of trade, microsimulation, automatic stabilisers, public finances

JEL codes: F13, F14, F16, D31, H23, I38

* **Acknowledgements:** I thank Max Ghenis and Seyed Mahdi Hosseini for valuable comments and helpful discussions. This is a working paper presenting first-pass public-data results; the household-microsimulation stage is specified in Section 7.

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1 Introduction

Britain has just lived through the densest sequence of trade shocks in its post-war history: three acts of trade policy—a new customs border with its largest trading partner (2021), a tariff war with its largest single-country export market (2025), and a new trade agreement with India (2026)—plus the sharpest trade-*transmitted* shock since the 1970s, the 2022 terms-of-trade deterioration, and one near-zero agreement (CPTPP, 2024) that serves as a benchmark. The policy/market distinction is carried explicitly throughout: the energy episode is not a policy act, and the comparison never treats it as one. Each episode has its own literature; none has been studied at the level where trade shocks are actually felt—the household, through the tax–benefit system. This paper studies all of them there, together, under one lens.

The design holds the second stage fixed and varies the shock. Each episode enters as a *declared first stage*—published estimates for the negative shocks, the government's own projections for the positive ones, never re-estimated here—and is mapped onto households through two channels: consumer prices against expenditure patterns, and labour income through the statutory tax–benefit system. Because gross sizes span two orders of magnitude, comparisons run in ratios: burden as a share of spending by income decile, cushioning shares, and per-pound-of-shock intensities. And because the rulebook itself moved across the window—the Covid uplift removed in 2021, upratings lagging a year behind prices, the Energy Price Guarantee legislated mid-shock—the design treats the tax–benefit vintage as an explicit dimension: the same shock is scored under the rules of its day, under the previous year's rules, and with the discretionary layer switched off.

The first-pass results already discipline the received wisdom in three ways. First, the negative shocks were regressive and the positive ones were not progressive: the gross energy shock put 12.39 per cent of bottom-decile spending at stake against 5.77 per cent at the top (fixed baskets, before the policy response), the TCA food-price effect 0.35 against 0.20 per cent on the window-average path, while the India agreement's consumer gains are proportional and worth pounds-per-year, not per cent. Second, each episode's regressivity is generated by its category's budget-share gradient: food takes 14.9 per cent of bottom-decile spending against 8.3 at the top, energy 7.0 against 3.2, clothing and footwear roughly equal shares everywhere—composition, not size, decides who bears a trade shock. Third—the finding the rulebook axis exists for—the British state cushioned its trade shocks with different instruments on different margins: the earnings-led tariff shock was absorbed automatically (36.6 to 43.9 per cent, from the companion study), the price-led energy shock was absorbed by discretion (the Energy Price Guarantee took 46.2 per cent of the gross shock), and the automatic price-side mechanism—indexation—ran a year late, costing a Universal Credit household £393 (9.8 per cent of the allowance) in the crisis year, months after a £1,040 uplift had been withdrawn.

The contribution is threefold. First, comparability: we are not aware of any study, for any country, that places multiple trade shocks under a common statutory household lens; the closest traditions—the World Bank's tariff incidence framework, the CGE-microsimulation linkages, the EUROMOD shock literature—each supply one ingredient (Section 2), and this paper's first pass seeds the join that its specified microsimulation stage completes. Second, the rulebook axis: holding a shock fixed and varying the vintage of the welfare state that receives it has, to our knowledge, no precedent, and post-Covid Britain—four shocks landing on four rulebooks—is its natural laboratory. Third, honesty about epistemic status as a design principle: every first stage is labelled estimate or projection, every dial is declared, and one episode (CPTPP) is carried as a near-zero benchmark whose only defensible output—an aggregate divided by a household count—is reported as exactly that.

The paper is a working draft in the same staged convention as its companion (Ahmadi, 2026): the first pass runs the declared first stages against published expenditure-by-decile tables and statutory arithmetic (Sections 5–6); the household-record stage—PolicyEngine UK on the enhanced Family Resources Survey, whose consumption imputation and parameter history have been verified for exactly this use—is specified in Section 7. Section 3 defines the estimand, Section 4 the five episodes and their declared first stages, and Section 8 concludes.

2 Literature

Four literatures meet here without previously having met each other: the household incidence of trade policy, the microsimulation of shocks through statutory tax–benefit systems, the distributional analysis of price shocks and benefit indexation, and the UK's episode-specific empirical record since 2020. Each has done part of what this paper does; none has joined them.

2.1 Trade to households

The framework for pushing trade shocks onto household microdata is best developed for developing countries: the World Bank's Household Impacts of Tariffs programme simulates tariff changes through both expenditure and income portfolios for 54 countries (Artuç et al., 2021, 2019), and the OECD has published the concordance machinery for mapping CGE trade-model output onto household-budget-survey categories (Luu et al., 2020). The structural anchor for the consumption side is Fajgelbaum and Khandelwal (2016): trade's gains flow disproportionately to poorer households through traded-goods budget shares. The 2025 tariff war produced a fast-moving US applied literature—the Budget Lab's rolling distributional scoring of the enacted schedules and the Tax Policy Center's implementation of tariffs inside a working tax microsimulation model (Budget Lab, 2025; TPC, 2025)—and a first academic treatment of two-sided incidence on household expenditure microdata (Fleck and Pradhan, 2026). The CGE–microsimulation linkage tradition supplies the architecture for putting a trade model upstairs and microdata downstairs (Robilliard and Robinson, 2005; Peichl, 2008; Colombo, 2010; Calì et al., 2019); of that tradition, only Peichl (2008) carries actual statutory tax rules in the micro layer.

What none of this work contains is a tax–benefit system responding to the shock. The World Bank and OECD frameworks stop at market incomes and expenditure; the US 2025 work is consumption-side; the CGE-linkage papers (Peichl's flat-tax application aside) use reduced-form income equations. As far as we can establish, no trade shock has been run through a statutory tax–benefit calculator— EUROMOD, UKMOD, TAXBEN, PolicyEngine or any counterpart—in any country, for any episode.

2.2 Shocks through statutory rules

That machinery exists and has been applied to every kind of shock except trade. Dolls et al. (2012) measure the stabilisation coefficients of the US and European systems against stylised income and unemployment shocks; Brewer and Tasseva (2021) impose the observed Covid labour-market shock on FRS microdata through UKMOD and decompose the cushioning into automatic and discretionary parts; the EUROMOD nowcasting tradition supplies the labour-transition machinery (Navicke et al., 2014). Two studies come close enough to this paper's rulebook axis that the claim must be stated narrowly. Amores et al. (2025) run the 2022 inflation shock and its offsetting fiscal measures through EUROMOD for the EU, separating income-support from price-containing instruments— in effect a price-shock stabilisation coefficient, and the direct antecedent of what this paper computes; they find fiscal measures recovered roughly a third of welfare losses while targeting them poorly. Popova et al. (2026) are closer still on the UK: UKMOD with LCFS consumption imputed onto the FRS, disposable and consumable income, and an explicit CPI-versus-earnings indexation counterfactual over 2019–2023. Neither models the September-CPI *timing* lag, the Energy Price Guarantee, or any trade shock, and neither treats automatic stabilisation in anything but the classic income-shock sense. That quartet—timing lag, discretionary energy layer, trade first stages, and the comparison across episodes—is what this paper adds, and it is a narrower claim than "nobody has measured price-shock stabilisation."

The conceptual gap it rests on is old and explicit: simulating an oil-price supply shock in FRB/US, Cohen and Follette (2000) found that automatic stabilisers do very little against a supply shock, unlike their demand-shock performance—a result never taken to microdata. Paulus et al. (2020) supply the canonical treatment of non-indexation as a distributional phenomenon, and Leventi et al. (2024) the nearest existing treatment of indexation itself as a stabiliser. The companion study (Ahmadi, 2026) is, to our knowledge, the first tax–benefit microsimulation of a trade shock anywhere—a single episode (the 2025 US tariffs), earnings channel only. This paper generalises it: four episodes, both channels, and the rulebook vintage as an explicit dimension.

2.3 Price shocks, indexation and the rulebook

The paper's rulebook axis has more precedent than its trade axis, and the debts should be stated plainly. Cribb et al. (2023) quantified the 2022–23 uprating shortfall directly—benefits rose 3.1 per cent against roughly ten per cent inflation, recipients were not made whole until April 2025—and scored the flat cost-of-living payments against a timely-indexation counterfactual, finding the discretionary route cost more while leaving hundreds of thousands of means-tested households worse off than indexation would have. OBR (2022b) puts the aggregate real fall at around five per cent, some £12 billion, in 2022–23. Judge and Murphy (2023) supply the fact that pre-empts the obvious objection that lags wash out over a cycle: across fifteen years, working-age benefits were uprated below outturn inflation in ten fiscal years, so the asymmetry is empirical, not assumed. Internationally, Eckerstorfer et al. (2025) microsimulate Austria's inflation episode with lagged wage and pension adjustment, the closest non-UK analogue. This paper adds neither the discovery of the lag nor a better estimate of it; what it adds is the join—the same lag priced against *trade-shock* incidence, so that the timing of indexation and the import content of a household's basket are evaluated in one place, across episodes rather than one.

One contrary claim must be met rather than cited in passing. Jaravel (2021) concludes that trade affects price levels roughly similarly across income groups, innovation rather than trade driving inflation inequality. The episodes here are consistent with that as a statement about trade *in general* and inconsistent with it as a statement about these shocks in particular: what makes the UK's negative episodes regressive is not trade exposure as such but the categories they landed on—food and domestic energy, whose budget shares fall steeply with income (Figure 3). Where a trade shock lands on clothing, as the India agreement does, this paper finds exactly the flat incidence Jaravel describes.

The method template for the consumption channel predates all of this: Cravino and Levchenko (2017) showed after Mexico's 1994 devaluation that poorer households' cost of living rose about half again as much as richer households', because they spend more on tradeables and buy cheaper varieties within categories—the second mechanism being one that category-level data cannot see. The decomposition machinery for price shocks is also established: Sologon et al. (2025) decompose inflation's welfare and distributional impact across expenditure patterns, and Chen et al. (2024) map an exchange-rate shock onto heterogeneous household price experiences using scanner data. Their finding is a direct bound on this paper's first pass: inflation inequality surged in 2021–23 partly because cheaper goods inflated fastest, a within-category quality margin that published expenditure-by-decile tables hold fixed and therefore understate. On the policy question the episodes raise, the sharpest numbers are Italian: Curci et al. (2025) find targeted social bonuses costing a fifth as much as untargeted VAT and excise cuts drove most of the inequality reduction, with nearly half of untargeted energy support accruing above the median.

On the price-response side, Levell et al. (2025) is the frontier and a standing challenge to any decile-based exercise: using transaction data they find income explains only a small share of the variation in energy spending, so within-decile dispersion in exposure dwarfs the between-decile gradient, and they show that an optimal package retains a strictly positive price subsidy even when transfers can condition on income and past usage. That is a principled defence of a price-capping instrument, and it disciplines what this paper's proportional-cushioning arithmetic can be taken to mean. Bonnet et al. (2025) reach the opposite conclusion for transport fuel in France, and the disagreement is commodity-driven—fuel budget shares rise with income where domestic-energy shares fall—which is precisely the kind of composition effect the present framework is built to make visible. Auclert et al. (2023) add the externality a domestic incidence exercise cannot see: national price subsidies insulate one importer at the cost of world energy demand. Cross-country, Ari et al. (2022) and Amaglobeli et al. (2026) document that most European crisis support was untargeted and price-distorting, while NAO (2024) records the fiscal counterpart in the UK—an initial £139 billion estimate against a £44 billion outturn, the option value of a cap that binds only in bad states.

2.4 The UK episode record

Each episode carries its own first-stage literature, reviewed with the imported numbers in Section 4: the Brexit/TCA price and trade effects (Bakker et al., 2026; Breinlich et al., 2022; Freeman et al., 2022), the 2022 terms-of-trade and energy-price shock and its policy response, the 2025 US tariff schedule (OBR, 2025), and the India agreement's impact assessment (DBT, 2025). The Brexit episode has the richest distributional record, and two precedents sit close to this paper. Breinlich et al. (2016) asked literally who would bear the pain, distributing projected Brexit price changes across income groups and household types, and Clarke et al. (2017) did the same for an MFN tariff scenario, finding the largest hits on unemployed, single-parent and pensioner households. Both are ex-ante scenario exercises predating the actual border; this paper's contribution relative to them is realised episodes, a statutory second stage, and comparison across shocks. Davenport and Levell (2021) supply the earnings-side counterpart (barrier exposure by occupation, sector and region), and Alabrese et al. (2026) and Dhingra et al. (2026) the 2026 regional accounting—the latter with new TCA tariff and non-tariff-measure estimates in a spatial general-equilibrium model, finding the damage runs through real wages and productivity rather than structural change.

A near-namesake deserves flagging: Pallotti et al. (2023) ask who bore the costs of the euro-area inflation shock and find consumption losses roughly flat across the income distribution, with the incidence falling instead along an age gradient as rigid rents hedged poorer households.

That the same question yields a vertical answer in Britain and an age-based one in the euro area is itself a reason to hold the lens fixed and vary the shock.

The record also contains a puzzle this paper is built to address. Breinlich et al. (2022) find the depreciation episode's consumer cost roughly *flat* across the income distribution; Bakker et al. (2026) find the non-tariff-barrier episode sharply *regressive*, the bottom decile's cost-of-living increase around half again larger than the top's. Same country, adjacent years, opposite incidence—because the two shocks landed on different baskets. A framework that holds the household lens fixed and varies the shock is the instrument that settles such disagreements, and Section 6 shows the same mechanism operating across five episodes. What remains absent everywhere is the tax–benefit response: the Brexit price literature never asks what the benefit system gave back; the energy-crisis literature scores the discretionary response but not the trade shock as such; the FTA impact assessments contain, in DBT's own regulator's words, household analysis "developed further for future FTA IAs." The gap this paper fills is the join: the same statutory second stage, run against all four first stages.

3 Framework: one estimand, four shocks, two channels, six rulebooks

3.1 The estimand

For every episode the object is the same conditional quantity: the statutory household response to a declared trade-shock first stage. Let a first stage specify (i) a vector of consumer-price changes by consumption category, and/or (ii) a labour-income loss or gain with a sectoral location and an adjustment margin. The second stage maps that first stage onto households and computes the change in disposable income, its distribution, and the Exchequer's share, under the tax–benefit rules in force at the episode date. The headline metric is the *cushioning rate*—one minus the household disposable-income change per pound of gross shock, the income-stabilisation coefficient of Dolls et al. (2012) generalised to price shocks by measuring the gross shock as the expenditure-weighted cost imposed at fixed consumption—because ratios, unlike levels, are comparable across episodes whose gross sizes span two orders of magnitude.

Nothing in the design estimates the first stages. Each is imported from published estimates and labelled with its source and its attribution caveats (Section 4); the paper's contribution begins where those estimates stop, at the household and the Exchequer. This is the conditional-estimand convention of the companion tariff study (Ahmadi, 2026), applied retrospectively—with the advantage that retrospective first stages can be anchored to realised outturns rather than ex-ante bridges.

3.2 Two channels

Trade shocks reach households through prices and through earnings, and the post-Covid episodes split cleanly on that line. The consumption channel prices a category-level price vector against household expenditure shares: implemented at the household level on imputed consumption (the microsimulation stage) and, in this paper's first-pass results, on published expenditure-by-decile tables. The earnings channel imposes sectoral labour-income changes on employees by industry and runs them through the statutory system, with the adjustment margin (wage cuts versus displacement) as a declared scenario dimension—the machinery, conventions and caveats of the companion study carry over unchanged. The channels are reported separately and never summed without an explicit interaction note: a price shock erodes real income that the benefit system partially indexes, while an earnings shock triggers means-tested entitlement; the two responses run through different statutory mechanisms, and their comparison is the point.

3.3 The rulebook axis

The same shock cushions differently depending on when it lands, because the rulebook moves: benefit uprating follows lagged inflation, so a fast price shock outruns indexation for a year; thresholds frozen in cash terms tighten in real terms through an inflation episode; and discretionary interventions (most dramatically the Energy Price Guarantee) rewrite the effective schedule mid-shock. The design therefore separates three quantities for each episode: the cushioning delivered by the rules *as they stood*, the cushioning the same rules would have delivered under contemporaneous full indexation (the uprating-lag cost), and the discretionary layer on top. Price-shock incidence has been run through tax–benefit models before (Section 2), and indexation counterfactuals are standard in that literature; what has not been done is to hold a *trade* shock fixed and vary the rulebook vintage that receives it, or to separate the statutory timing lag from the discretionary layer within a single episode. The post-Covid sequence—four shocks landing on four different vintages of the same welfare state—is a natural laboratory for exactly that comparison.

3.4 What a decile lens can and cannot see

Two recent results bound what any distribution-by-decile exercise, including this paper's first pass, is entitled to claim. Borusyak and Jaravel (2024) show that for trade shocks the overwhelming share of the variance of household welfare changes lies *within* income deciles rather than between them, because exposure is driven by product-level consumption patterns that income predicts weakly; Levell et al. (2025) reach the same conclusion for energy on UK transaction data. The implication is not that decile gradients are wrong—they are real, and they are what Section 6 measures—but that they are a small part of the dispersion in who was hurt. Decile means answer "was this shock regressive?"; they do not answer "who was hurt?", and the two questions have different answers. This is the strongest single argument for the household-record stage of Section 7, where exposure and entitlement are observed jointly at the household level, and it is why the first-pass results are reported as gradients rather than as welfare statements.

A second bound concerns channels. Decomposing inflation's household impact, Ferreira et al. (2026) find the net-nominal-position and labour-income channels an order of magnitude larger than the consumption-basket channel. The basket channel is the entire content of the price-side exercise here: savers and borrowers, mortgagors and depositors, are not in this paper's ledger, and the earnings channel enters only for the one episode where the companion study supplies it. The estimates are therefore a lower bound on the distributional consequences of these episodes, and are labelled as such throughout.

3.5 What the design is not

The estimates are first-round statutory incidence, not general equilibrium: no behavioural response, no wage or employment feedback from the price shocks, no substitution in consumption (the price-channel numbers are first-order, envelope-theorem approximations to welfare changes and upper bounds on avoidable cost). Episode first stages carry their own literatures' uncertainty, which the design brackets by importing low/central/high variants where the literature

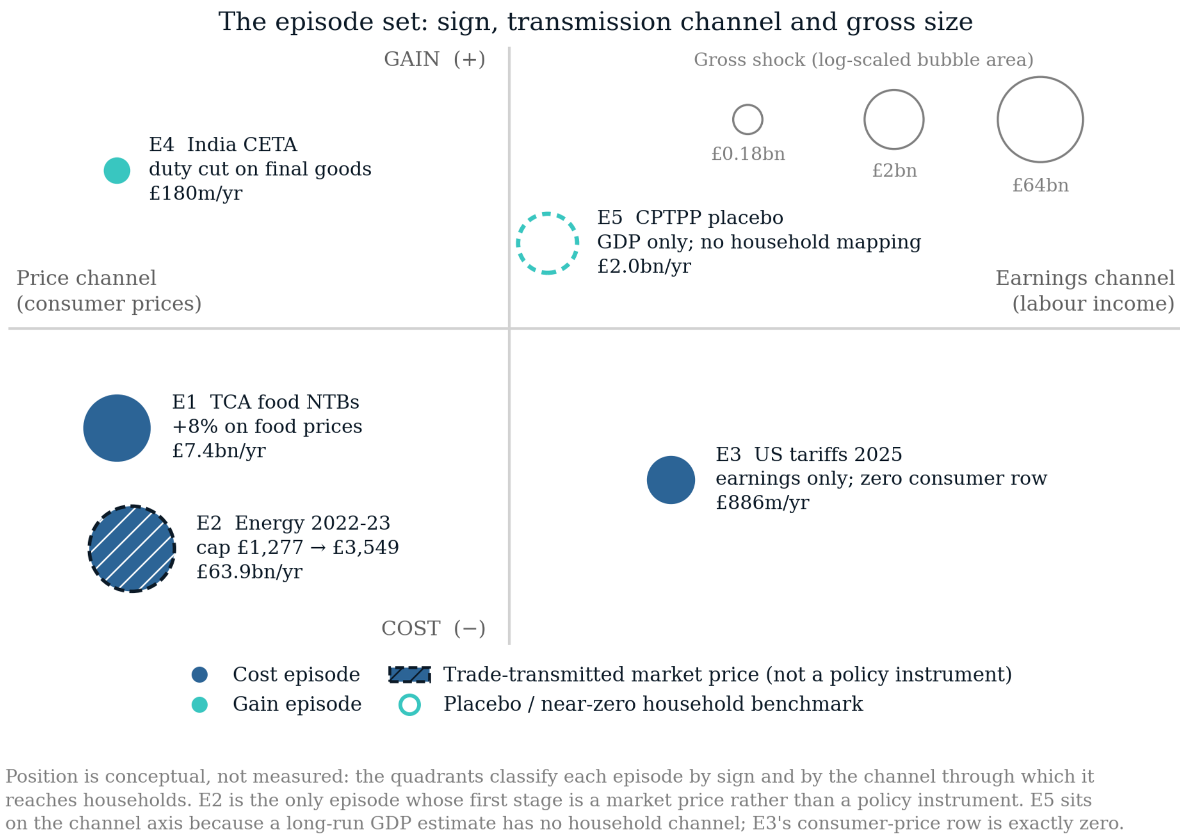


Figure 1: The episode set. Horizontal: the channel through which the shock reaches households; vertical: cost or gain. Bubble area is log-scaled in the gross shock. The 2022 energy episode (hatched, dashed ring) is trade-*transmitted* rather than a policy act; CPTPP (open, dashed) is the near-zero benchmark.

provides them and labelling single-point stages as such. The comparison across episodes is a comparison of conditional scenarios that share a second stage, not a causal decomposition of realised household outcomes over 2021–2026, which would require separating the trade shocks from everything else that happened—a task this design deliberately declines.

4 Five episodes and their declared first stages

Each episode enters the analysis as a declared first stage: published estimates, imported with citations and attribution caveats, never re-estimated. The five span the design's axes: negative and positive, price-led and earnings-led, measured and projected, policy-made and trade-transmitted. One is carried as a near-zero benchmark. Figure 1 places the set on the paper's two axes and Table 1 lists each declared first stage with its source and epistemic status.

4.1 Episode 1: the TCA border (2021–23), negative, price-led

The Trade and Cooperation Agreement took effect in January 2021, replacing single-market membership with a zero-tariff but non-tariff-barrier-laden border. The price evidence is the

Table 1: Declared first stages. Every price or earnings vector is an imported published estimate; none is re-estimated in this paper.

Episode	Date	Type	Channel	First-stage source	Key imported number	Status
E1 TCA food NTBs	Dec 2019–Mar 2023	Policy	Consumer prices (food)	Bakker et al. (2026)	+8% on food prices (6% low variant)	Estimate (ex post)
E2 Energy 2022–23	Oct 2022–Mar 2023	Market	Consumer prices (energy)	Ofgem (2022); OBR (2022)	Cap £1,277 → £3,549; EPG £2,500	Observed (statutory)
E3 US tariffs 2025	2025	Policy (foreign)	Earnings only; no UK retaliation	Ahmadi (2026)	£886m/yr gross earnings shock	Estimate (projection)
E4 India CETA	2026+	Policy	Consumer prices (clothing, footwear, food)	DBT (2025)	£180m/yr duty cut on final goods	Projection (ex ante)
E5 CPTPP benchmark	Long run	Policy	Aggregate GDP; no household mapping	DBT (2023)	+£2.0bn GDP (+0.08%)	Projection (long run)

Notes. Type distinguishes a policy instrument from a trade-transmitted market price. Source entries are short cite keys; see the bibliography. E3's consumer-price row is exactly zero (the UK imposed no retaliatory tariffs), so its household incidence runs through earnings alone.

cleanest of any episode: exploiting cross-product variation in EU import intensity, Bakker et al. (2026) attribute roughly 8 percentage points of the 25 per cent food price rise between December 2019 and March 2023 to Brexit non-tariff barriers—a cumulative £250 per household on food alone, £6.95 billion in aggregate, with pass-through of NTB costs to consumers of 50–80 per cent and the largest effects in high-EU-import products such as meat and dairy (around 10 points above low-exposure comparators). The trade-volume record is equally stark: relative UK imports from the EU fell about 25 per cent on entry (Freeman et al., 2022), and by 2024 goods exports were an estimated £27 billion (6.4 per cent) below counterfactual, with some 16,400 small firms exiting EU exporting entirely (Freeman et al., 2024). The earnings side, by contrast, rests on assumptions rather than estimates—the OBR's standing 15 per cent trade-intensity and 4 per cent productivity wedges (OBR, 2026)— so this episode's earnings channel is declared as an assumption-based scenario and its price channel as an estimate-based one. Attribution caveat: the price identification is relative (cross-product), so economy-wide effects may be missed, and the counterfactual is contested by critics of the OBR's trade assumptions.

4.2 Episode 2: the terms-of-trade shock (2022–23), negative, price-led, very large

The 2022 energy and import-price surge was, in the Bank of England's description, a terms-of-trade shock of a size unseen since the mid-1970s: the terms of trade deteriorated 6.2 per cent in the year to 2022Q3 (ONS, 2023), import prices rose 20 per cent relative to output prices over two years (Broadbent, 2022), the energy import bill roughly doubled to £117 billion in 2022 (IMF, 2023)—the *increase* of roughly £60 billion being a transfer abroad of about 2.5 per cent of GDP (derived arithmetic, labelled as such)— and of the roughly 17 per cent CPI rise to 2022Q4, more than 70 per cent (12.7 points) was imported (Dhingra and Page, 2023). The household-level price vector is policy-mediated and well documented: the typical annual energy bill's cap rose from about £1,277 (winter 2021–22) to £3,549 (October 2022), truncated to £2,500 by the Energy Price Guarantee, at a discretionary cost the OBR put at £27 billion for the EPG within £47 billion of price subsidies and £78 billion of total energy support (OBR, 2022). Counterfactual no-policy welfare losses average 6 per cent of after-tax income, reaching 11 per cent at the 95th percentile of exposure (Levell et al., 2025), against energy budget shares of roughly 11 per cent in the second income decile versus 6 per cent in the ninth (Resolution Foundation, 2022). This is the strongest first stage of the five—multiple independent official estimates of both the aggregate shock and the retail price vector, plus a published imported/domestic decomposition—and the design decision it forces is constitutive for this paper: the shock is declared *gross*, and the Energy Price Guarantee is modelled as what it was, a discretionary rulebook intervention, so that automatic and discretionary cushioning can be separated.

4.3 Episode 3: the US tariffs (2025–26), negative, earnings-led

The April 2025 US schedule (10 per cent baseline, 25 per cent autos and steel) and its partial mitigation are the subject of the companion study (Ahmadi, 2026), which supplies this episode's earnings first stage and its second-stage results: a tariff-calibrated earnings loss of roughly £886 million a year, cushioned at 43.9 per cent when delivered as broad wage cuts and 36.6 per cent as displacement. The external anchors are the OBR's March 2025 scenarios (import-demand elasticity 0.4; peak GDP effects of −0.6 to −1.0 per cent across scenarios) (OBR, 2025) and the Bank of England's effective-tariff estimates (11 per cent at April 2025, about 9 per cent after the Economic Prosperity Deal) (Bank of England, 2025). The UK consumer side of this episode is, to date, literally zero: the UK has not retaliated, so no tariff-induced consumer price vector exists—an empirical fact the comparative design records rather than an omission.

4.4 Episode 4: the India agreement (2026–), positive, both channels

The UK–India Comprehensive Economic and Trade Agreement entered into force on 15 July 2026. Its first stage is declared from the government's own impact assessment (DBT, 2025): long-run GDP $+£4.8$ billion a year (+0.13 per cent by 2040), real wages +0.19 per cent ($\approx £2.2$ billion), and—on a different clock, at entry into force—consumer duty reductions of £220 million a year, £180 million of it on final goods, concentrated in clothing (£132 million), footwear (£13 million) and food and beverages (£12 million). Sectoral GVA effects give the earnings channel its geography: machinery and equipment +1.65 per cent, textiles and apparel −0.68 per cent. Two dials are declared rather than estimated, following the IA's own caveats: preference utilisation (the duty figures assume 100 per cent on static 2022 trade) and consumer pass-through of the duty savings. The honest framing—and the reason a government projection can anchor a conditional design—is that the exercise computes what the government's own central estimates imply for households, which its impact assessment does not report and its regulatory scrutineer asked for (RPC, 2026).

4.5 Episode 5: CPTPP accession (2024–), the near-zero benchmark

The UK's accession to the CPTPP (in force December 2024) carries a central government estimate of $+£2.0$ billion a year in the long run—0.08 per cent of GDP, about £70 per household per year before any incidence structure—because the UK already had agreements with nine of the eleven members (DBT, 2023). It is included precisely because it is nearly nothing: a data-limitation benchmark whose honest treatment—an aggregate divided by a household count, and no more—marks the boundary of what a declared first stage can support.

4.6 What the five episodes jointly reveal about the record

One pattern runs across the dossier and should be stated carefully, because its naive version is mechanical: past shocks naturally have ex-post estimates and new agreements ex-ante projections. The non-mechanical fact is on the evaluation side—no UK trade agreement has ever been ex-post evaluated at the household level, although CPTPP has now been in force long enough for the exercise, and the India assessment's own regulatory scrutineer flagged the consumer analysis as underdeveloped (RPC, 2026). The comparative design keeps the epistemic split explicit throughout—estimate-based and projection-based first stages are labelled in every table—and the episode set is typed policy/market, with the terms-of-trade episode carried as a trade-transmitted market shock, never as a policy act.

5 First-pass method

The results in this working draft run the declared first stages against public aggregates; the household-record stage is specified in Section 7. Everything is generated by one pipeline script (`incidence.py`) that downloads and pins its raw inputs and emits every reported value as a machine-generated macro; no number is typed by hand except the declared first-stage constants of Section 4, which are typed *with their citations* and marked as imported.

Following the companion studies' convention, the epistemic status of every input is stated once:

Object	Status
Episode first stages	**Imported** from cited published estimates (episodes 1–2) or government projections (episodes 4–5); episode 3 from the companion study. Never re-estimated here.
Expenditure by category × decile	**Data** (ONS Family Spending detailed workbooks).
Energy price-cap levels, statutory benefit rates, CPI outturns	**Data** (Ofgem/DESNZ, DWP, ONS).
Pass-through and utilisation dials	**Assumed**, bracketed, entering the pipeline once each.
Incidence arithmetic	**Computed**; no estimation step anywhere in the pipeline.

Consumption incidence. Each episode's price vector is applied to the relevant categories of household expenditure by equivalised-disposable-income decile, giving a per-decile burden or gain in pounds a year, as a share of total spending, and as a share of disposable income. These are first-order calculations: no substitution, no behavioural response, fixed baskets—upper bounds on avoidable welfare cost, in the envelope-theorem sense standard in this literature. Where a published aggregate exists (the £250 cumulative food cost of the TCA episode; the OBR's Energy Price Guarantee cost), the pipeline reports its own aggregate beside it as a cross-check, not a calibration.

Rulebook arithmetic. The vintage axis is seeded with statutory arithmetic on public parameters: the real-terms shortfall of the Universal Credit standard allowance during the 2022–23 inflation episode relative to contemporaneous indexation (the uprating-lag cost, computed month by month against CPI outturns), and the discrete removal of the £20-a-week uplift in October 2021—the rulebook event that immediately preceded the terms-of-trade shock. The Energy Price Guarantee enters as the discretionary layer of episode 2: the gap between the gross price shock and the EPG-truncated shock is the discretionary cushioning share, computable per decile from cap levels alone.

Two data caveats. The FYE2025 Family Spending workbook carries an ONS correction notice affecting percentage standard errors across the high-level COICOP categories; the point estimates this paper uses are unaffected, but no sampling-uncertainty statement is made on the expenditure inputs for that reason. And the two vintages used (FYE2022 for the TCA and energy episodes, whose spending bases must predate the shocks, and FYE2025 for the India agreement) sit either side of an ONS change in the equivalence scale, so income levels are not comparable across vintages—which is why every cross-episode comparison in this paper runs on shares of spending rather than shares of income.

Comparability. Episodes are compared in ratios, not levels: burden as a share of spending by decile, cushioning shares where computable, and a per-£-billion-of-shock normalisation, because gross sizes span more than two orders of magnitude and pound totals would reduce the comparison to a statement about the energy episode. Scenario dials (pass-through, utilisation) are reported as labelled variants, never blended into central cells; brackets are outer bounds, not confidence intervals; and estimate-based versus projection-based first stages are flagged in every table.

6 First-pass results

6.1 The comparative anatomy

Table 2 is the paper's central object: the five episodes side by side under the common lens. Three regularities organise everything that follows. First, *the negative shocks were regressive and the positive ones were not progressive*: the gross energy shock would have cost the bottom decile 12.39 per cent of its spending against 5.77 per cent at the top (before the policy response and demand adjustment), the TCA food shock 0.35 against 0.20 per cent of spending on the window-average path (1.19 against 0.66 at end-state), while the India agreement's consumer gains are spread almost exactly proportionally (0.015 versus 0.015 per cent of spending) and are two orders of magnitude smaller. Second, *regressivity follows the affected category's budget-share gradient*: food takes 14.9 per cent of bottom-decile spending against 8.3 at the top, energy 7.0 against 3.2, while clothing and footwear—where the CETA gains land—are flat across the distribution: the composition of a shock, not just its size, determines who bears it. Third, *the cushioning came from different parts of the state*: the one earnings-led episode was cushioned automatically (36.6–43.9 per cent through the tax–benefit system, from the companion study), while the price-led energy episode was cushioned by discretion—the Energy Price Guarantee absorbed 46.2 per cent of the gross shock—and the automatic price-side mechanism, benefit indexation, arrived a year late (Section 6.3).

Table 2: Five shocks under one lens: first-pass comparability

Episode	Type, channel	Gross shock	% of spending D1	% of spending D10	Cushioning
TCA food, 2021–23	policy, prices	£2.2 bn	0.35	0.20	in estimate
Energy, 2022–23	market, prices	£63.9 bn	12.39	5.77	EPG 46.2%
US tariffs, 2025–26	policy, earnings	£886 m	zero		auto. 36.6–43.9%
India CETA, 2026–	policy, prices	£180 m[g]	0.015	0.015	n/a
CPTPP, 2024–	benchmark	£2.0 bn[g]	no structure		n/a

Notes: [g] gain, not cost. Deciles are of equivalised disposable income; entries are first-order incidence on published expenditure-by-decile tables at fixed baskets. TCA on the window-average path (end-state variant in the text); energy gross of the policy response and demand adjustment. Episodes are typed policy/market: the energy episode is trade-*transmitted*, not a policy act. "Zero" records that the UK did not retaliate, so no consumer price vector exists. Cushioning entries measure different mechanisms (a pass-through convention, a price subsidy, a tax–benefit stabilisation coefficient) and are not summable; the common cushioning estimand of Section 3 is computed at the microsimulation stage, not here. Category budget-share gradients, which generate these incidence patterns, are in Figure 3; full per-decile detail in Table 3.

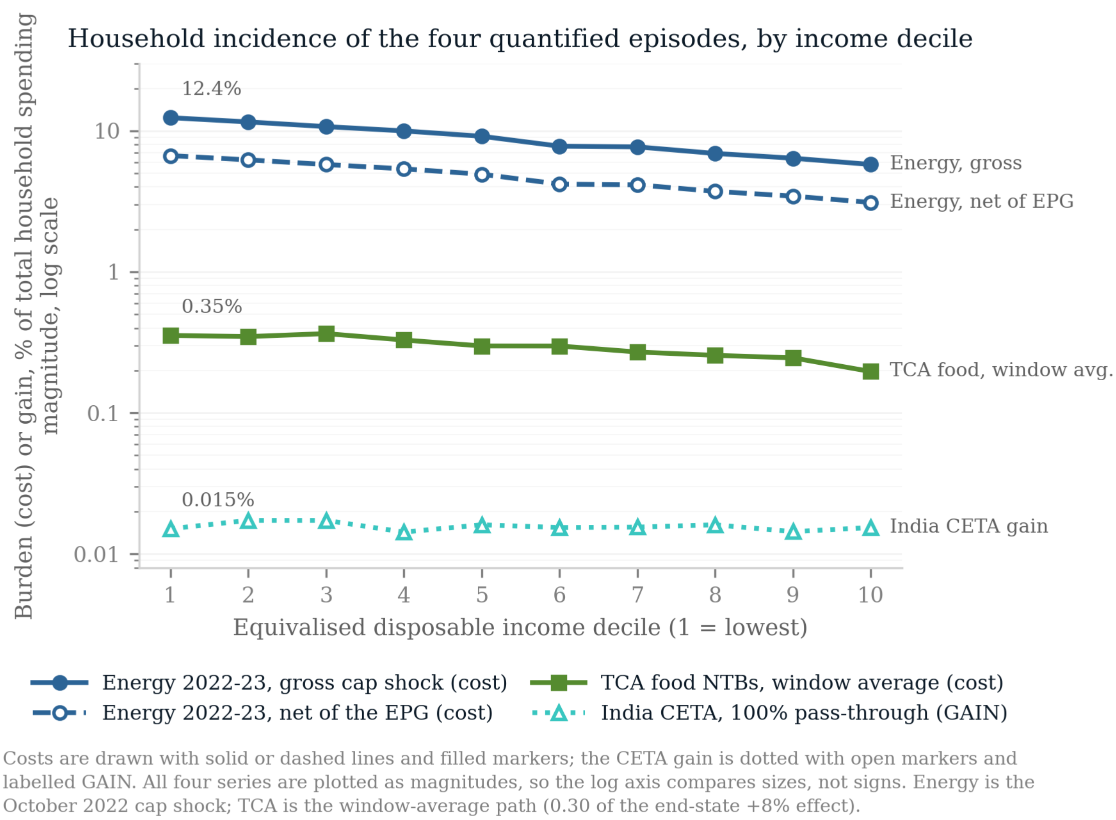
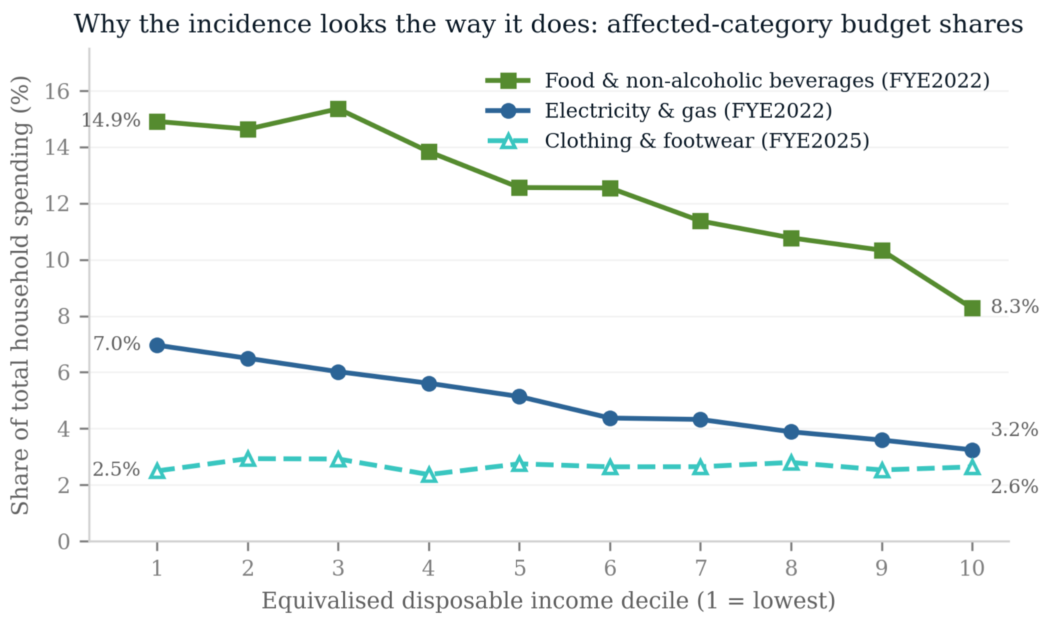


Figure 2: Household incidence by equivalised disposable income decile, as a share of each decile's total spending. Costs are plotted as positive; the India CETA series is a gain (dotted, open markers). The energy episode is shown gross and net of the Energy Price Guarantee.



Food and energy shares fall steeply with income, which is what makes those two episodes regressive; the clothing & footwear share is flat, which is what makes the CETA gain roughly proportional. Each share is computed on its own episode's spending vintage (ONS Family Spending 3.1E), so levels are not strictly comparable across the two vintages.

Figure 3: Why each episode has the incidence it does: the affected categories' shares of total spending by decile. Food and domestic energy fall steeply with income; clothing and footwear—where the India CETA gains land—are flat.

Table 3: Household incidence of the quantified episodes, by equivalised disposable income decile. TCA food NTBs are on the window-average path (0.30 of the end-state +8% effect); energy is the 2022–23 cap shock, gross and net of the Energy Price Guarantee; the India CETA column is the consumer gain at 100% pass-through. Costs are positive.

	TCA food NTBs		Energy, gross		Energy, net of EPG		CETA gain
Decile	£/yr	% spend	£/yr	% spend	£/yr	% spend	£/yr
1	58	0.35	2,017	12.39	1,086	6.67	2.96
2	63	0.35	2,082	11.56	1,120	6.22	3.93
3	70	0.37	2,045	10.71	1,101	5.77	4.47
4	74	0.33	2,257	9.98	1,215	5.37	3.90
5	72	0.30	2,202	9.14	1,185	4.92	5.12
6	84	0.30	2,193	7.78	1,180	4.19	5.37
7	80	0.27	2,285	7.69	1,230	4.14	5.61
8	84	0.26	2,257	6.91	1,215	3.72	6.56
9	90	0.25	2,350	6.39	1,265	3.44	6.90
10	94	0.20	2,748	5.77	1,479	3.11	9.93
D1/D10 ratio	—	1.80	—	2.15	—	2.15	0.98†

Notes. £ per household per year; % spend is the burden or gain as a percentage of that decile's total expenditure. The EPG cushions 46.2% of the gross energy shock at every decile by construction. † CETA ratio is on the %-of-spending basis (£ gains are not comparable across deciles). Spending bases: ONS Family Spending workbook 1, sheet 3.1E, FYE2022 vintage (TCA, energy) and FYE2025 vintage (CETA).

6.2 Episode detail

TCA food prices. The headline incidence uses the window-average path implied by the published cumulative: the £250-per-household estimate implies an average effect of about 0.30 of the 8 per cent end-state over the window, giving £58 (bottom decile) to £94 (top decile) a year, £2.2 billion a year in aggregate— regressive as a share of spending (0.35 versus 0.20 per cent) because food takes 14.9 per cent of bottom-decile spending against 8.3 at the top. The end-state figures (£194–315 a year; £7.37 billion) are reported as a labelled variant: the pipeline's linear-ramp cross-check against the published cumulative comes in at ratio 1.68, showing the price effect was back-loaded, so end-state overstates the lived burden. The microsimulation stage will carry the published path.

The energy shock and the EPG. At cap levels, the gross price shock (£1,277 to £3,549 typical bill, +177.9 per cent) *would have* cost the bottom decile £2,017 a year at fixed baskets—12.39 per cent of its total spending, more than double the top decile's share. Three labelled qualifications bound that number. A demand-response variant (unit elasticity applied at half weight, consistent with the double-digit weather-adjusted demand reductions observed in 2022–23) scales the burden to 69 per cent of the fixed-basket figure (£1,389 for the bottom decile; £44.0 billion aggregate). An imported-component bound, applying the Dhingra and Page (2023) finding that over 70 per cent of the CPI rise was imported, caps the trade-transmitted share of the gross aggregate at roughly £44.7 billion; the remainder is domestic cost and margin. And the annualised run-rate holds the October 2022 cap for a full year, which no household experienced; the six-month window figure is £14.8 billion of EPG absorption against the OBR's £27 billion costing—a context anchor, not a validation. Within those bounds the shape is

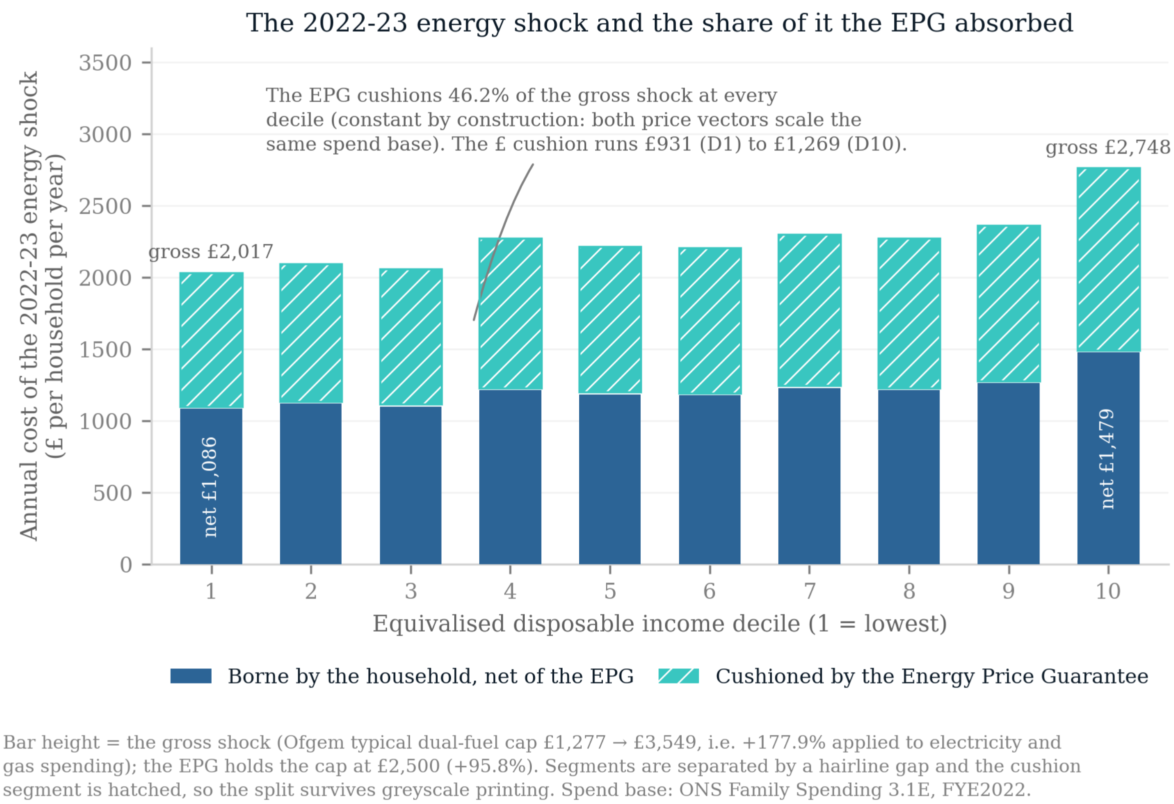


Figure 4: The 2022–23 energy shock by decile: the gross cap-level burden split into the part borne by households net of the Energy Price Guarantee and the part the Guarantee absorbed. Because the instrument capped a price, its cushioning share is constant across the distribution by construction.

robust: the Energy Price Guarantee truncated the rise to +95.8 per cent, absorbing £931 a year for a bottom-decile household and 46.2 per cent of the gross shock. Because the EPG capped the *price*, its cushioning share is constant across deciles by construction. Two qualifications follow from the frontier evidence. First, decile means understate the problem the instrument faced: Levell et al. (2025) show that income explains only a small share of the variation in household energy spending, so exposure within a decile is far more dispersed than the gradient between deciles—an argument for the household-record stage, where that dispersion is observable, and against reading any decile-mean cushioning share as a statement about individual households. Second, a proportional instrument is not thereby a mistaken one: Levell et al. (2025) find that an optimal package retains a strictly positive price subsidy even when transfers can condition on income and past usage, precisely because income targets energy exposure so poorly. The finding here is descriptive—the cap cushioned proportionally—not a welfare verdict. Whether the overall discretionary response was progressive cannot be judged from the EPG alone: the Energy Bills Support Scheme's flat £400, the means-tested cost-of-living payments and the council-tax rebate— jointly of the same order as the EPG and deliberately targeted—are not modelled in this first pass, and the distributional verdict on the discretionary stack is deferred to the microsimulation stage.

The US tariffs. The UK consumer side is zero—no retaliation occurred—so this episode's incidence is entirely earnings-led: a gross loss of £886 million a year cushioned by the automatic tax–benefit system at 36.6 per cent (displacement) to 43.9 per cent (wage cuts), the companion study's headline carried over as this episode's second stage.

The India gains, and the placebo. At full pass-through the CETA duty cuts are worth £2.96 a year to a bottom-decile household and £9.93 to a top-decile one—roughly proportional as a share of spending, though pro-rich in pounds (the top decile gains about three times as much in cash), and economically negligible at any pass-through dial (£1.48–4.97 at 50 per cent). The CPTPP row is a near-zero benchmark, included as a data-limitation case rather than a placebo test: its £2.0 billion long-run GDP estimate divides to £70 per household per year with no defensible distributional structure, and the pipeline reports exactly that and nothing more.

6.3 The rulebook seed results

The vintage axis is seeded with two statutory facts. First, the uprating lag: Universal Credit's standard allowance rose 3.1 per cent in April 2022 (indexed to the previous September's inflation) while prices rose at double digits. Against month-by-month contemporaneous indexation—an upper-bound counterfactual no benefit system operates—a single adult's standard allowance fell £393 short over 2022–23 (9.8 per cent of the annual allowance); against the policy-feasible alternative of a single April 2022 uprating at contemporaneous annual CPI, £230. Both figures are for the single-adult standard allowance only, before elements and caseload weighting; the range, not either point, is the seed result. Neither the lag nor its cost is a discovery of this paper: Cribb et al. (2023) quantified the 2022–23 shortfall and scored the discretionary cost-of-living payments against a timely-indexation counterfactual, and OBR (2022b) put the aggregate real fall at roughly £12 billion—a figure the household-record stage should reproduce as a validation target. What is new here is the join: the lag priced against trade-shock incidence, on the same households, across episodes. The lag also does not wash out over the cycle: Judge and Murphy (2023) count ten of fifteen years in which working-age benefits rose by less than outturn inflation. The shortfall is lost to timing alone, at exactly the moment the terms-of-trade shock peaked. Figure 5 shows the gap month by month. Second, sequencing: the £1,040-a-year Covid uplift was removed in October 2021, months before the energy shock landed, so the lowest-income households entered the largest price-led trade shock of the period from a rulebook trough. Together with the EPG's absorption of 46.2 per cent of the gross price shock, the seed picture of the British response is: *automatic on the earnings margin, discretionary on the price margin, and lagged precisely where the price shocks bit hardest*—with the progressivity of the full discretionary stack an open question for the microsimulation stage. The microsimulation stage turns these representative-household facts into population-weighted decompositions.

7 The microsimulation stage

The first-pass results of Section 6 run the declared first stages against published expenditure-by-decile tables and representative-worker statutory arithmetic. The full stage runs them inside PolicyEngine UK on the enhanced Family Resources Survey, and four things change.

First, the consumption channel moves from decile tables to household records: the enhanced FRS carries imputed household consumption across the twelve COICOP divisions plus the sub-items that matter most for trade shocks (petrol and diesel separately, in pounds and litres;

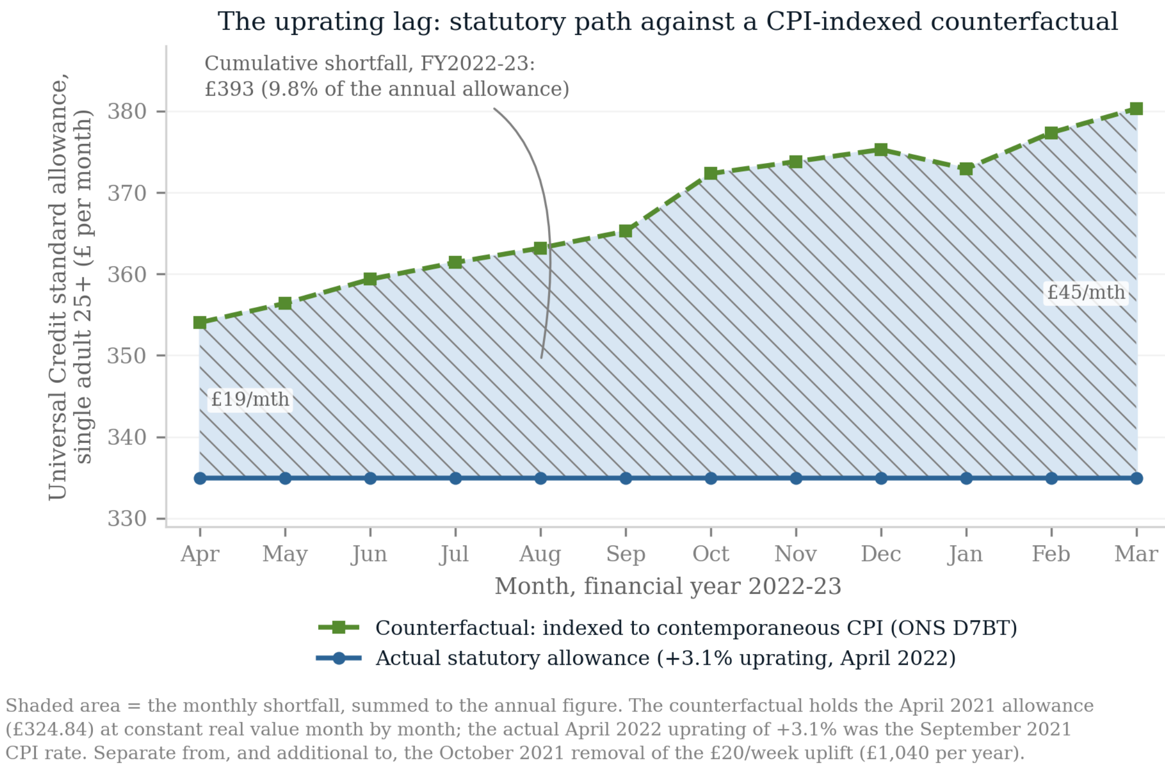


Figure 5: The uprating lag in the crisis year. The Universal Credit standard allowance for a single adult aged 25 or over (statutory path, flat from April 2022) against a counterfactual indexed to contemporaneous CPI. The shaded area is the shortfall, summing to £393 over 2022–23.

electricity and gas separately; bus fares), imputed by quantile regression forest from the LCFS 2023–24 and calibrated to administrative energy and fuel totals. Each episode's price vector therefore prices each household's own basket, jointly with that household's benefit entitlements—the covariance between exposure and cushioning that decile tables cannot see. Tariff-line and category-level first stages must be pre-aggregated to those roughly sixteen categories, and the imputation freezes budget-share *patterns* at the 2023–24 donor vintage across simulated years, a limitation the stage will state. Second, the earnings channels run through the companion study's displacement machinery on the same households, so the two channels of a given episode land on a common population and their incidence can be compared household-by-household rather than distribution-by- distribution. Third, the rulebook axis becomes exact: PolicyEngine's parameter history carries dated, legislatively cited values through the whole window (the Covid-era Universal Credit uplift and its October 2021 removal, the March 2022 fuel-duty cut, the 2022 and 2023 upratings), its own test suite exercises the 2020–2023 rule years, and counterfactual-rulebook runs—an earlier year's parameters applied to a later survey vintage—are native to the architecture. Each episode is therefore scored under the rules in force at its date, under the preceding year's rules (isolating uprating lag), and with the discretionary interventions switched off (isolating the automatic component)—the three-way decomposition of Section 3 computed rather than approximated. Two verified limits shape the earlier years: no 2021-vintage enhanced-FRS data year exists (the earliest survey base is 2022–23), so 2021 rulebooks are applied to later households with incomes re-deflated; and the survey's Universal Credit/legacy-benefit split follows reported receipt rather than the historical rollout, so 2021–22 rule-year results understate legacy-benefit interactions and are restricted to the tax and UC-parameter channels. Fourth, the poverty and inequality outputs gain their correct household equivalisation and the Exchequer effects their full statutory coverage.

The build order follows the data dependencies. The consumption-channel implementation extends PolicyEngine's existing consumption variables (used today for carbon and VAT analysis) with the episode price vectors—a parameter reform per episode, no new imputation. The earnings channels are re-runs of existing machinery with episode calibrations. The rulebook runs require only parameter-year switches plus explicit flags for the discretionary layers (the Energy Price Guarantee being the largest). The licensed inputs are the same FRS/EFRS files the companion study uses; nothing else is gated.

Two methodological upgrades are available at the same time and are specified here. First, the fixed-weight decile indices of the first pass can be replaced by non-homothetic distributional cost-of-living indices, which Hochmuth et al. (2025) construct from exactly the inputs this project already has—aggregate prices, expenditure shares and a single cross-sectional expenditure distribution—answering the standard objection that fixed baskets misprice large relative-price movements. Second, and more consequentially, the household-record stage is the level at which the within-decile dispersion documented by Borusyak and Jaravel (2024) and Levell et al. (2025) becomes visible: the same machinery that computes a decile gradient computes the full distribution of exposure within it, which is where most of the welfare variance lives.

Two honest limits will survive the full stage. The consumption imputation carries its own error, inherited from the LCFS donor models, and imputation error correlated with benefit status would bias the exposure-cushioning covariance; the stage will report the imputation diagnostics alongside the results. And the rulebook decomposition attributes to "discretion" whatever the parameter history records as legislated change, which bundles genuine shock response with policy that would have happened anyway—a labelling choice, stated as such, not an identification claim.

8 Discussion

Three findings survive every caveat the first pass carries. The negative shocks of the 2020s—two policy-made, one trade-transmitted—were regressive through prices, and the positive policy shocks offer no offsetting progressivity: partly because they are small, partly because their gains arrive in categories (clothing, footwear) whose budget shares are flat across the distribution. The state's cushioning was margin-specific: automatic where the shock hit earnings, discretionary where it hit prices—and the largest discretionary instrument, capping a price, cushioned proportionally by construction, though whether the discretionary *stack* (with its flat and means-tested payments alongside the cap) was progressive is deferred to the microsimulation stage. Proportional is not the same as mistaken: because income predicts energy exposure poorly, a positive price subsidy survives in the optimal package (Levell et al., 2025), and the cap's fiscal cost was itself state-contingent—an initial £139 billion estimate against a £44 billion outturn (NAO, 2024). The instrument's defensibility is commodity-specific: where budget shares rise with income, as for transport fuel, transfers dominate subsidies (Bonnet et al., 2025); the Guarantee applied to the commodity where the case for a price instrument is strongest. What a domestic incidence exercise cannot price at all is the externality that national subsidies impose through world energy demand (Auclert et al., 2023). And the automatic price-side stabiliser the welfare state does possess—indexation—is calibrated to a world where prices move slowly: against a fast terms-of-trade shock it delivered its help a year late, which for the poorest households was the difference between the rulebook cushioning the shock and amplifying it.

One institutional fact deserves stating precisely, because its naive version is a truism: that past shocks have ex-post estimates and recent agreements ex-ante projections is mechanical. What is not mechanical is that the United Kingdom has never ex-post validated a trade agreement's impact assessment at the household level—CPTPP has been in force long enough for the exercise, and its absence is a fact about evaluation practice that the Regulatory Policy Committee's criticism of the India assessment's consumer analysis makes concrete (RPC, 2026). On the government's own projections, the positive policy episodes are small and distributionally flat; on the published estimates, the negative episodes are large and regressive. Whether that ledger balances is exactly the question a household-level evaluation practice would answer, and no one keeps the books.

The limits are those declared throughout. First stages are imported, with their identification and their controversies; the incidence is first-order and static; the channels are not summed; the rulebook decomposition labels legislated change as discretion without asking what would have been legislated anyway. The first pass runs on decile tables and representative households, not on household records—the covariance between exposure and entitlement, the poverty and inequality effects, and the population-weighted rulebook decompositions all await the microsimulation stage, which the verification work reported in Section 7 shows is buildable with existing tools on existing data.

What the framework offers, finally, is a standing capability rather than a one-off study: every future trade shock—a retaliation schedule, the next agreement, the next terms-of-trade event—arrives as one more declared first stage for the same second stage. A country that intends to conduct an active trade policy while running a means-tested welfare state needs exactly this join, and as of this writing no institution, in Britain or elsewhere, maintains it.

Data and code availability

The first-stage evidence dossier, the costing/incidence pipeline, pinned raw downloads and generated result macros are available in the accompanying GitHub repository (`https://github.com/PolicyEngine/uk-trade-shock-study`, directory `analysis_multishock/`); an archived release with a persistent identifier will accompany publication. All inputs to the first-pass results are public (ONS expenditure and price statistics, published first-stage estimates, statutory parameters). The microsimulation stage additionally requires the licensed Family Resources Survey via the UK Data Service, which is not redistributed. Episode first stages are imported from cited published estimates and are not re-estimated here.

Declarations

The author reports no conflicts of interest. No external funding was received for this study. The project uses secondary, aggregate public statistics and published estimates; no new human-participant data were collected. The author is responsible for the analysis, interpretation, and any remaining errors.

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